Question Paper

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MANIPAL ACADEMY OF HIGHER EDUCATION

Discrete Mathematical Structures [MAT 2101]

Marks: 30 **Duration: 90 mins.**

MCQ

Answer all the questions. Section Duration: 20 mins

1) Which ONE of the following is a monoid but not a group... (0.5)

The set of all non-singular real matrices of order 2 under "matrix addition".

The set of all integers under "subtraction".

The set of all subsets of a non-empty set under the set operation "union of sets".

The set of all real square matrices of order 2 under "matrix addition".

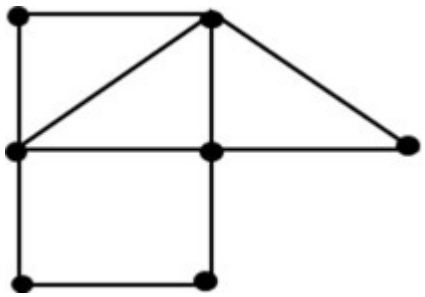
2) Which ONE of the following is NOT a subgroup of a group $\{1, -1, i, -i\}$ with respect to multiplication? (0.5)

$\{i, -i\}$ $\{1\}$ $\{1, -1\}$ $\{1, -1, i, -i\}$

3) Singular values of the matrix $\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$ are... (0.5)

1, 9 0, $\sqrt{3}$ 1, 3 1, $\sqrt{3}$

4) Radius of the following graph is... (0.5)



2 3 4 7

5) Given a graph G = (V, E), the square graph G^2 is defined as the graph with the same vertex set $V(G^2) = V(G)$, where two vertices $u, v \in V$ are adjacent in G^2 whenever $d(u, v) \leq 2$ in G. If $G = C_4$ (cycle of length 4), then G^2 is isomorphic to... (0.5)

C_4 K_4 (Complete graph with 4 vertices) $K_{2,2}$ (complete bi-partite graph) Path P_4

6) Let G={1,3,5,7} be a group under the operation "multiplication modulo 8". Then the inverses of the elements 3, 5 and 7 respectively, are (0.5)

3, 5 and 7 | 5, 3 and 7 | 5, 7 and 3 | 7, 5 and 3

7) Let H = 3ℤ (set of all integers which are multiples of 3) and K = 5ℤ (set of all integers which are multiples of 5) be subgroups of (ℤ, +). Which ONE of the following is NOT a subgroup of ℤ? (0.5)

H ∩ K | H + K = {h + k : h ∈ H, k ∈ K} | H ∪ K | {0}

8) If the number of closed walks of length 3 in a graph G is 24, then the number of triangles in G is… (0.5)

2 | 4 | 6 | 8

9) Let G be a graph with 21 edges. Suppose 3 vertices of G are of degree 4 and the remaining vertices are of degree 2. Then how many vertices G has? (0.5)

18 | 10 | 16 | 9

10) If $A=\begin{bmatrix}1&1&0\\1&0&1\\0&1&1\end{bmatrix}=\begin{bmatrix}1&0&0\\l_{21}&1&0\\l_{31}&l_{32}&1\end{bmatrix}\begin{bmatrix}u_{11}&u_{12}&u_{13}\\0&u_{22}&u_{23}\\0&0&u_{33}\end{bmatrix}$ then l_{21}=... (0.5)

1 | 0 | 2 | -1

Descriptive Questions

Answer all the questions.

11) Find a singular value decomposition of the matrix $A=\begin{bmatrix}1&0&0\\0&2&0\end{bmatrix}$. (4)

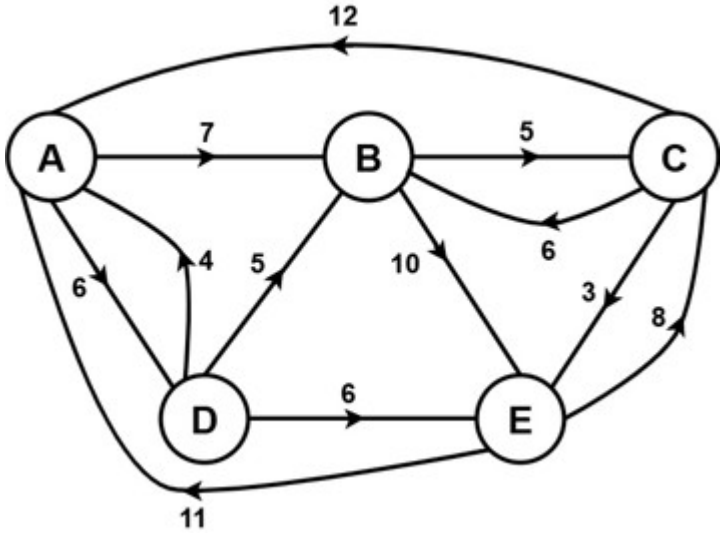
12) Solve the following system of equations by using LU decomposition method. (3)

$$x+y+z=1$$

$$4x+3y-z=6$$

$$3x+5y+3z=4$$

13) Use Dijkstra's algorithm to find the shortest path from B to all other vertices of G in the following network. (3)

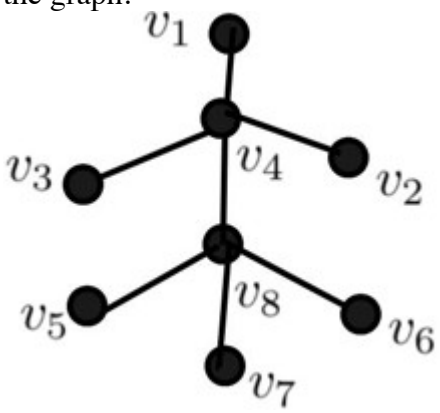


14) Let $*$ be defined on the set of integers as $x * y = x + y - 2, \forall\, x, y \in \mathbb{Z}$. Then verify whether $(\mathbb{Z}, *)$ is a group. Justify your answer. Is $(\mathbb{Z}, *)$ Abelian? (3)

15) i). Let (G, *) be a group and H be a nonempty subset of G. If $a * b^{-1} \in H$ for all $a, b \in H$ then show that H is a subgroup of G.

ii). Let G = $\mathbb{Z}_8$ be a group under the operation 'addition modulo 8'. Construct the cayley table for the subset H={0, 2, 4, 6} of G under 'addition modulo 8'. Is H a subgroup of G? Justify your answer. (3)

16) If G is a connected bi-partite graph with at least 3 vertices, then show that G has no odd cycles. Is the following graph bi-partite? If so, write down the partite sets of the graph.



(3)

17) Show that the number of vertices in a self-complementary graph G is of the form 4k or 4k+1, where k is a positive integer. (2)

18) Show that if G is a group and $a^2 = e$ for all $a \in G$, then G is Abelian. (2)

19)	A tree T with 50 pendant vertices (Vertices of Degree 1) has an equal number of vertices of degree 2,3,4, and 5 and no vertices of degree greater than 5. Find the number of vertices in T.	(2)

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